Understanding Unix

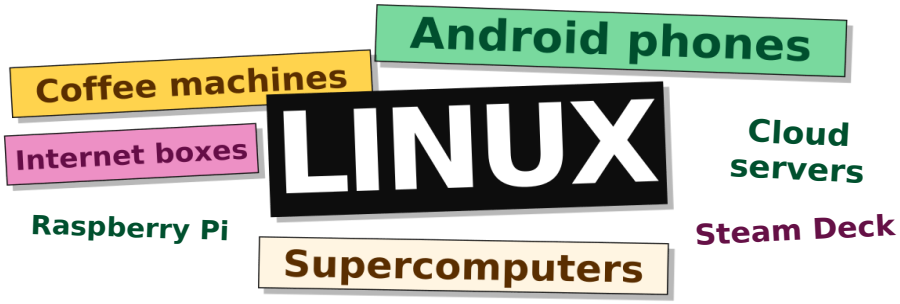
NUnix · L3 MIAGE · CM1 · 90 minutes

ISTIC · University of Rennes 2026–2027

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Linux is closer than you think

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You can use it all day without opening a terminal.

Yes, even the coffee machine

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Android phones
The **Linux kernel** manages processes, memory and hardware.
Google · platform architecture

Freebox Server Pop
A home internet box. Free publishes its **Linux source code**.
Free · firmware 4.9.10

MARO Model One
A coffee machine with **embedded Linux**, remote support and updates.
Toradex · May 2026

Linux can run behind a touchscreen or a web interface.

Unix and Linux usage

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Mobile web use
69.17%
of observed mobile page views come from **Android**, built on Linux.
Statcounter · Sept 2026

Websites
92.1%
use the **Unix family**, including Linux, among sites whose OS W3Techs identifies.
W3Techs · 1 Oct 2026

Supercomputers
500/500
systems in the **June 2026 TOP500 list** use the Linux family.
TOP500 · June 2026

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$ whoami

Who teaches NUnix, and how is the course organised?

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Associate Professor
University of Rennes · ISTIC
IRISA / Inria · Taran

- Arithmetic circuits: **MLIR (LLVM) / CIRCT**.
- **Systolic arrays** and small floating-point formats.

Previously: Inria / INSA Lyon, Emeraude.
PhD: **BSC / UPC**.
bynaryman.github.io



MareNostrum 4 · BSC, Barcelona

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CM, TD and TP

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NUnix · L3 MIAGE · 2026–2027

Sessions	Teaching team
CM · lectures	Louis Ledoux
TD · tutorials	Elisabeth Bouriah and Louis Ledoux
TP · practical sessions	Elisabeth Bouriah and Louis Ledoux

In TD and TP, we practise the lecture topics through exercises, commands and Bash scripts.

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Course structure

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Three lectures (CM) of 90 minutes.

Lecture	Topics	Application
CM1 · Unix	Files, paths and permissions	Find files and check access
CM2 · Commands	Processes, search, pipes and redirections	Count errors in a log
CM3 · Bash	Variables, conditions, loops and functions	Write scripts with arguments and loops

TD and TP: practise commands first, then write scripts.

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One line, a useful result

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Which errors occur most often in **1,000 log entries**?

```
sort < errors.log | uniq -c | sort -nr | tee output.txt |
  a2ps -1 -o - | ps2pdf - rapport.pdf
```

Save the counts in `output.txt` and the report in `rapport.pdf`.

errors.log · first three lines

```
Database timeout
Login rejected
Connection refused
```

output.txt · first three counts

```
    320 Database timeout
    210 Login rejected
    140 Connection refused
```

`a2ps`: text → PostScript
`ps2pdf`: PostScript → PDF

By the end of this course, you will understand this command.

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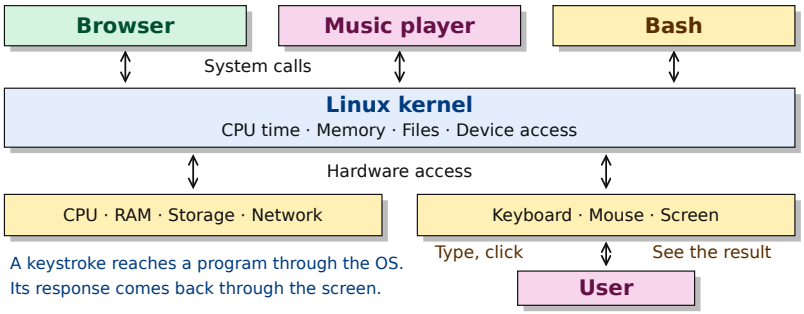
1. Linux and the shell

What happens between typing a command and the machine doing the work?

Programs, kernel and hardware

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We use hardware to interact with programs.



The kernel **shares resources and controls access**.

Linux

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- **1991:** Linus Torvalds announces his kernel project.
- Linux is a **Unix-like kernel**, distributed as free and open source software.
- A distribution combines the kernel with tools, libraries and applications.

Our working environment: a Linux system with the **Bash shell** and command-line tools.

Unix, GNU and Linux

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Name	Meaning
Unix	The operating-system family that inspired Linux
GNU	The project behind Bash and many command-line tools
Linux	The kernel: manages processes, memory and hardware

GNU = GNU's Not Unix.
The name contains itself: a recursive acronym.

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Touch, click or type

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You already use interfaces to start programs.

Your action	Interface	Example
Touch a phone screen	Graphical interface	Open an app from its launcher
Click with a mouse	Graphical interface	Open a folder in a file manager
Type a command	Text interface	Ask Bash to list a directory

A smartphone can be a **terminal**: an endpoint for using a service.
Its touchscreen, like a mouse or keyboard, provides **input**.

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Our terminal for this course?

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DEC VT100 · IF-3-SYS, G. Salagnac, INSA Lyon

A **physical terminal** sends keystrokes to a computer and displays its replies.
The programs run on the connected computer.

Good news: yours fits in a window.

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Terminal, shell and command

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Name	Role
Terminal emulator	A window for keyboard input and text output
Bash, the shell	Interprets the line and runs commands
`ls`, a command	Lists directory entries

Type `ls` → Bash runs `ls` → the terminal displays its output.

TTY = *teletypewriter*: early terminals printed on paper.
Unix keeps the name, even for emulated terminals.

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A command and its arguments

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```
ls -l /home/alice
```

Part	Role
`ls`	**Command**: list directory entries
`-l`	**Option**: change how `ls` lists them
`/home/alice`	**Operand**: the directory to list

Both `-l` and `/home/alice` are **arguments** passed to `ls`.

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Short and long options

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Form	Example	Meaning for GNU `ls`
Short: **one dash**	`ls -a`	Include hidden entries
Long: **two dashes**	`ls --all`	Same as `-a`
Combined	`ls -la`	`-l` and `-a` together

Some options take a value: `head -n 5 notes.txt` prints five lines.

A **bare `--` ends options**.
`ls -- -notes.txt` treats `-notes.txt` as a filename.

Each command defines its options. Check `--help` or `man`.

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Getting help

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```
man ls
info coreutils
apropos directory
help cd
```

- **Tab** completes a name or displays possible completions.
- **Up** recalls a previous command. `history` lists history.
- `clear` clears the display. In `man`, **q** quits.

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2. Files and directories

Where is my file, and what can I do with it?

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One directory tree

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A **directory** is a folder: it contains names of files and other directories.

```
/
├── home/
│   └── alice/
│       └── notes.txt
└── etc/
```

The common root is `/`. The path `/home/alice/notes.txt` follows the tree.
Alice's **home directory** is `/home/alice`; the university may use a different path.

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Personal files and system data

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Linux separates personal files, configuration and application data.

Directory	Typical contents
`/home/alice`	Alice's personal files
`/etc`	System configuration
`/var`	Changing data: logs, caches, application state
`/var/lib`	Persistent application state
`/var/log`	Log files

`/var/lib` and `/var/log` are subdirectories of `/var`.

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Users, administrators and software

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Directory	Typical role
`/home`, `/root`	User homes; the root account's home
`/usr`	Most distribution-provided tools and libraries
`/usr/local`	Software installed locally by the administrator
`/opt`	Add-on software packages

`/` is the filesystem root.
`root` is also the administrator account's name.

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Temporary files and system information

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Directory	Purpose / lifetime
`/var`	Changing data, often persistent
`/tmp`	Temporary files; cleanup policy varies
`/run`	Runtime state, cleared at boot
`/dev`	Device interfaces
`/proc`, `/sys`	Views of processes, kernel and devices

A path can expose live system information.
It need not refer to ordinary data stored on disk.

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Several filesystems, one tree

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```
/
├── home/       user files
├── mnt/        a temporary mount point
└── media/      removable media mount points
```

Mounting makes another filesystem visible at a directory.
The device node might be `/dev/sdb1`; its files appear at the mount point.

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The current directory

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Notation	Meaning
`.`	Current working directory
`..`	Parent directory
`/`	Root directory
`~`	Your home directory, expanded by the shell
`$HOME`	Variable containing your home path
`$PWD`	Variable containing the current directory path

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Absolute and relative paths

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Current directory: `/home/alice`

Path	Interpretation
`/home/alice/notes.txt`	Absolute: start at `/`
`notes.txt`	Relative: start in the current directory
`./notes.txt`	The same relative path
`../bob/notes.txt`	Go to the parent, then into `bob`

A path tells us **where to look**. Permissions determine whether access is allowed.

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Moving around

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```
pwd
cd /home/alice
cd ..
cd
```

- `pwd` prints the current working directory.
- `cd path` changes the current directory.
- `cd` or `cd ~` returns to your home.

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Listing directory entries

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```
ls
ls -l
ls -a
ls -la /home/alice
```

- `-l` adds details, including ownership and permissions.
- `-a` includes names beginning with `.`, such as `.bashrc`.
- `ls directory` lists that directory's contents.

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Creating and removing directories

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```
mkdir practice
rmdir practice
```

`rmdir` removes an **empty** directory.

`rm -r practice` also removes the directory's contents.
Shell deletion does **not** send files to a recycle bin.

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Creating, copying and moving files

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```
touch notes.txt
cp notes.txt notes-copy.txt
mv notes-copy.txt archive.txt
rm archive.txt
```

`touch` creates a missing file, or updates timestamps of an existing file.

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Reading text files

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Command	Result
`cat notes.txt`	Print the whole file
`less notes.txt`	Browse interactively, **q** to quit
`more notes.txt`	Browse a page at a time
`head -n 5 notes.txt`	Print the first five lines
`tail -n 5 notes.txt`	Print the last five lines
`diff old.txt new.txt`	Show differences between two files

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Why is less called less?

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Mark Nudelman was investigating errors in large logs.
He wanted to read the lines **before** an error.

His version of `more` could not move backward.
The new pager's name joked about a **backward more**.

```
less app.log
```

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Concatenating files

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```
cat first.txt second.txt
cat
cat first.txt - second.txt
```

- With filenames, `cat` prints their contents **in argument order**.
- Without filenames, `cat` reads **standard input**.
- A standalone `-` means **read standard input here**.
- At an empty terminal input line, **Ctrl+D** indicates end of input.

The third command reads `first.txt`, then your keyboard input, then `second.txt`.

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A space can change the arguments

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We want to read **one file** named `meeting notes.txt`.

Without quotes

```
cat meeting notes.txt
```

`cat` receives **two filenames**:

- `meeting`
- `notes.txt`

With quotes

```
cat "meeting notes.txt"
```

`cat` receives **one filename**:

- `meeting notes.txt`

Quotes keep the filename together. The surrounding `"` characters are **not passed to `cat`**.

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Three ways to protect a space

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Each command passes the same filename: **`meeting notes.txt`**.

Form	What you type
Single quotes	`cat 'meeting notes.txt'`
Double quotes	`cat "meeting notes.txt"`
Backslash	`cat meeting\ notes.txt`

The surrounding quotes and the backslash protect the space. They are **not part of the filename**.

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Patterns select matching filenames

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The directory contains only **a.txt** and **meeting notes.txt**.

```
cat *.txt
```

The shell finds the matches, then passes **two filenames** to cat: a.txt and meeting notes.txt.

Pattern	Matches
*	Zero or more characters
?	Exactly one character
[abc]	One character from the set

The space **inside a matched filename** stays intact.

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Quotes keep a pattern literal

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Same directory: **a.txt** and **meeting notes.txt**.

Let the shell find matches

```
cat *.txt
```

cat receives **two filenames**:

- a.txt
- meeting notes.txt

It reads both files.

Protect the star with quotes

```
cat '*.txt'
```

cat receives **one filename**:

- *.txt

No file has that literal name here: cat reports an error.

cat "*.txt" and cat *.txt also pass a literal *.

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Single quotes or double quotes?

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HOME stores your home directory, here /home/alice. $HOME asks for its value; echo prints its arguments.

What you type	What you see
echo '$HOME'	$HOME
echo "$HOME"	/home/alice

Single quotes keep $HOME literal. **Double quotes** allow it to expand. Both protect spaces and filename pattern characters such as *.

With double quotes, we can use the variable and protect the space:

```
cat "$HOME/meeting notes.txt"
```

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Checkpoint: files and paths

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1. What does cd .. do?
2. Which ls option includes hidden entries?
3. What does cat a.txt b.txt print first?
4. Which filename does cat '*.txt' try to open?

1. Move to the parent directory.
2. -a
3. The contents of a.txt.
4. The literal filename *.txt. The shell does not expand the star.

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3. Who may access this file?

Why can I read this file, but cannot change or delete it?

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Users, groups and ownership

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```
whoami
id
ls -ld .
ls -l
```

- `id` shows your user identity and groups.
- `ls -l` shows the file's owner, group and permission bits.
- `ls -ld` inspects a directory itself.

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Permission classes

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```
-rwxr-xr--
  u  g  o
```

Class	Applies to
u	The file's owner
g	Members of the file's group
o	Other users
a	All three classes when using `chmod`

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What r, w and x allow

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Bit	Regular file	Directory
r	Read contents	List entry names
w	Modify contents	Add or remove entries*
x	Execute	Search / traverse

*Changing directory entries also requires search permission. Other restrictions, such as the sticky bit, may apply.

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Changing permission bits

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```
chmod u+x script.sh
chmod go-w notes.txt
chmod u=rw,g=r,o= notes.txt
```

- `+` adds the selected permission bits.
- `-` removes them.
- `=` sets the selected class to the listed permissions.

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Numeric permissions

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Each class adds its bit values: **r = 4**, **w = 2**, **x = 1**.

Owner	Group	Others
`rwx` = **7**	`rw-` = **6**	`r--` = **4**

```
chmod 764 notes.txt
```

The three digits describe **owner, group, others**, in that order.

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Checkpoint: permissions

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1. Decode `640` for a regular file.
2. Add execute permission for the owner of `run.sh`.
3. Which directory bit allows traversal?

1. Owner: `rw-`
   Group: `r--`
   Others: `---`
2. `chmod u+x run.sh`
3. `x`

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“I can see the name, but cannot read it”

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```
# Illustrative server metadata
 drwxr----- alice students  shared/
 -rw-r----- alice students  shared/notes.txt
```

Bob belongs to `students`. He can list the names, but cannot open `notes.txt`. Why?

The directory’s group bits lack **x**: traversal.
If team access is intended, the owner can add `g+x`.

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CM1 summary: files and permissions

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1. Locate configuration, software, data and logs.
2. List and read files; quote names containing spaces.
3. Check owner, group and other permissions.

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Optional reference

Examples to revisit during TD or personal practice

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Operating system, shell and GUI

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Layer	What it gives us
Operating system	Processes, storage and access control
Graphical interface	Windows, menus and direct visual interaction
Shell	Commands we can combine and save in scripts
Application	A task-specific tool, such as Excel or a database

Your laptop can run Windows while the company’s application runs on Linux. You need to work with both.

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Summarise a log before opening it in Excel

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Quantity	Meaning
2,000,000 log lines	Example input size
1,048,576 rows	The maximum in one Excel worksheet
One row per error code	The summary the team wants to discuss

Process the raw log where it lives, save the counts, then use Excel to analyse or present the summary.

Microsoft: worksheet limits. Power Query and the Data Model offer other ways to handle larger datasets.

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Path components and word counts

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```
dirname /home/alice/notes.txt
basename /home/alice/notes.txt
wc -w notes.txt
```

- `dirname` prints `/home/alice`.
- `basename` prints `notes.txt`.
- `wc -w` counts **words**. `wc -l` counts newline characters.

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Hard links and symbolic links

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Hard link

```
ln notes.txt other.txt
```

- Another name for the **same file**.
- Removing one name leaves the other usable.
- Same filesystem, normally no directory links.

Symbolic link

```
ln -s notes.txt shortcut
```

- Stores a **path** to a target.
- May cross filesystems or target a directory.
- Can become dangling if the target path disappears.

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Builtins and external commands

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Builtin

Implemented inside the shell.

```
cd ..
```

Can change the current shell's state.

External program

Loaded from an executable file.

```
ls -l
```

The shell searches directories in PATH.

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Recognising a command

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```
type cd
type ls
type -a echo
help cd
```

`type` reports how Bash resolves a name.
`help` documents its builtins.

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Seeing each argument with printf

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Only two files: **`a.txt`** and **`meeting notes.txt`**.

`printf '<%s>\n'` displays each following argument on its own line.

```
printf '<%s>\n' *.txt
```

```
<a.txt>
<meeting notes.txt>
```

Two filename arguments.

```
printf '<%s>\n' '*.txt'
```

```
<*.txt>
```

One literal argument.

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